Evaluating the Impact of Synthetic Disfluency Insertion on LLM Summarization

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Introduction

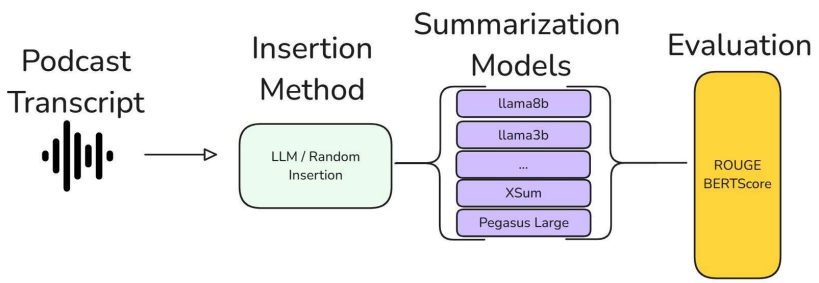
- LLMs have made large advancements to summarization quality in the past few years.
- While these models excel in written text, their performance on spoken-word language remains underexplored.
- **We evaluate how disfluent speech tokens, such as "um" and "uh," affect LLM summarization performance.**

Motivation

- Why? Traditional LLMs are predominantly trained on text data in which spoken word disfluencies are **underrepresented** [1]
- LLMs are increasingly being used in real-world speech applications such as Siri, ChatGPT Voice Mode

Experimental Framework

- In our experiments, we synthetically inject disfluencies **(low, medium, high, extreme levels)** into ~1k podcasts transcripts from **Spotify100k** [2] using gpt-4o-mini [3].
- We evaluate the summaries generated by different LLMs **{gpt-4o-mini, LLaMA, BART, T5, XSum, and Pegasus}** episode descriptions using **ROUGE (1, 2, L)** [4] **and BERTScores (Precision, Recall, F1)** [5]



Random Insertion

Algorithm 1 Random Disfluency Insertion

Require: Transcript T, mean μ, standard deviation σ (disfluencies per 1500 words)
Ensure: Transcript T' with inserted disfluencies

```
tokens ← tokenize(T)
W ← |tokens|
D_total ← count occurrences of "um", "uh" in tokens
r ~ N(μ, σ²); r ← max(0, r)
D_desired ← W × r/1500
p ← (D_desired − D_total)/W
T' ← []
for each token w in tokens do
    Append w to T'
    With probability p, append random choice from {"um", "uh"} to T'
end for
return T'
```

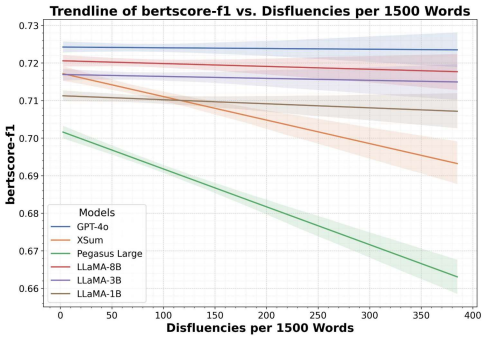
LLM Insertion

Prompt: [Please inject "um" or "uh" into the transcript at a {low; medium; high; extreme} level...]

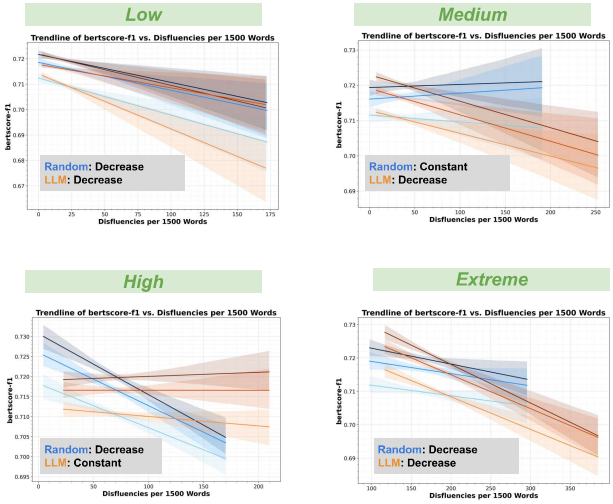
Generated Transcript:
[Welcome um, back to the podcast. Uh, today we will...]

RQ1: How does model size affect summarization performance?

- Across all levels, **higher parameter models are more robust** to disfluencies
- The trade-off in cost for larger models – with higher performance – may not be worth it



RQ2: How does the level of disfluency impact performance? [Preliminary Results]



Models
- Random-LLaMA-8b
- Random-LLaMA-3b
- Random-LLaMA-1b
- LLM-LLaMA-8b
- LLM-LLaMA-3B
- LLM-LLaMA-1B

- Overall, LLM **outperform** random insertions
- LLM perform better at **medium** level disfluency
- Random perform better at **high** level disfluency

References

[1] M Teleki, et al., "Comparing ASR Systems in the Context of Speech Disfluencies," in Interspeech, 2024.
[2] A. Clifton, et al., "100,000 Podcasts: A Spoken English Document Corpus," in COLING, 2020
[3] OpenAI, "ChatGPT," 2023. [Online]. Available: https://cdn. openai.com/papers/gpt-4.pdf
[4] C.-Y. Lin, "ROUGE: A Package for Automatic Evaluation of Summaries," in ACL, 2004
[5] T. Zhang, et al., "Bertscore: Evaluating text generation with bert," in ICLR 2020.

Future Work

Future work could investigate whether attention mechanism exhibits differential treatment of disfluencies based on position of the disfluency within a parse tree

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